

ASHISH RANJAN

+91 6202276736 | ashish.11ranjan01@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Final-year CS undergraduate with hands-on experience applying AI/ML, LLMs, and RAG frameworks (LangChain) to build scalable, API-driven applications. Proficient in Python and Java, with full-stack and cloud experience (AWS, Azure, OCI). Research experience includes a published paper on deep learning-based biomedical image segmentation.

EDUCATION

B.Tech, Computer Science Engineering (Cyber Physical Systems) Expected July 2027
Vellore Institute of Technology, Chennai CGPA: 9.03/10

TECHNICAL SKILLS

AI/ML: LangChain, RAG, LLMs, FAISS, PyTorch, Whisper, OpenCV, Streamlit
Programming: Python, Java, C, C++, JavaScript
Web & Backend: FastAPI, Node.js, Express.js, React.js, REST APIs, WebSockets, Microservices, HTTP/HTTPS, HTML, CSS
Cloud & Tools: AWS, Microsoft Azure, Oracle Cloud Infrastructure (OCI), Docker, Git, GitHub, VS Code
Databases: MongoDB, SQL, NoSQL, Mongoose, Data Model Design
Security: Cyber Security Fundamentals, IAM (Identity & Access Management), Cloud Security Methods
Operating Systems: Linux, Windows
Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP, Distributed Systems

PROJECTS

CogniFlow – Autonomous Agentic RAG & Live Context Engine [GitHub](#)

- Engineered a stateful, autonomous AI digital assistant using LangGraph to execute cyclic reasoning, dynamic multi-step planning, self-correction routing, and live external tool execution.
- Implemented a Dual Hierarchical Memory Engine utilizing SQLite for short-term session checkpoints and ChromaDB for durable, long-term semantic user fact retrieval.
- Integrated Model Context Protocol (MCP) for enterprise data extraction, streaming the agent's internal execution state to a React 18 UI via WebSockets with full LangSmith observability.

InterviewAI – AI-Powered Interview Preparation Platform [GitHub](#)

- Built a full-stack AI system using React, Node.js, FastAPI, MongoDB, and Docker with JWT authentication and resume-aware question generation.
- Implemented real-time WebSocket sessions with voice/text responses, speech-to-text, AI scoring, emotion analysis, and isolated code evaluation.
- Integrated LangChain, OpenAI/Gemini, FAISS, Whisper, and AI-driven performance reporting for personalized candidate analysis.

Resilio – AI-Powered Distributed System Reliability Platform [GitHub](#)

- Engineered an autonomous distributed-system reliability platform orchestrating containerized microservices deployed via Docker on a constrained Microsoft Azure virtual machine.
- Integrated automated chaos fault injection via the Docker API and real-time OpenTelemetry metric ingestion using Prometheus and WebSockets.
- Implemented a dual diagnosis engine featuring a local evidence-based heuristic analyzer and an LLM-driven root-cause analysis agent to detect and remediate cascading failures.

RESEARCH EXPERIENCE

Research Intern [2025](#)

Centre for Cyber Physical Systems (CCPS), VIT Chennai

- Developed deep learning models using Python and PyTorch for biomedical image segmentation.
- Automated training pipelines and optimized model performance through systematic debugging.
- Published research in Frontiers in Artificial Intelligence.

PUBLICATION

CA2PNet: A Context-Aware Multi-Scale Architecture with Adaptive Attention and Progressive Dilated Convolutions for Biomedical Image Segmentation

Frontiers in Artificial Intelligence (2026)

- Proposed a novel attention-based CNN architecture achieving 82.78% mIoU on BUSI and 85.15% mIoU on Kvasir-SEG.
- DOI: [10.3389/frai.2026.1802033](https://doi.org/10.3389/frai.2026.1802033)

CERTIFICATIONS

- Cyber Security – Basics and Application ([Verify](#)) – ISACA
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional ([Verify](#)) – Oracle
- The Complete Full-Stack Web Development Bootcamp ([Verify](#)) – Udemy